UR5e Robot Overview

The UR5e is a 6-degree-of-freedom collaborative robotic arm from Universal Robots. The complete system mainly consists of the robot arm, Control Box, and Teach Pendant.

1. Robot Arm
The robot arm consists of six rotating joints that provide the robot's movement. From the base to the end of the arm, these are:

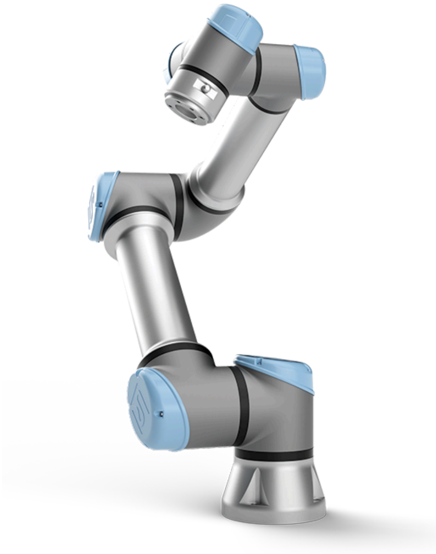
Base → Shoulder → Elbow → Wrist 1 → Wrist 2 → Wrist 3 → Tool Flange

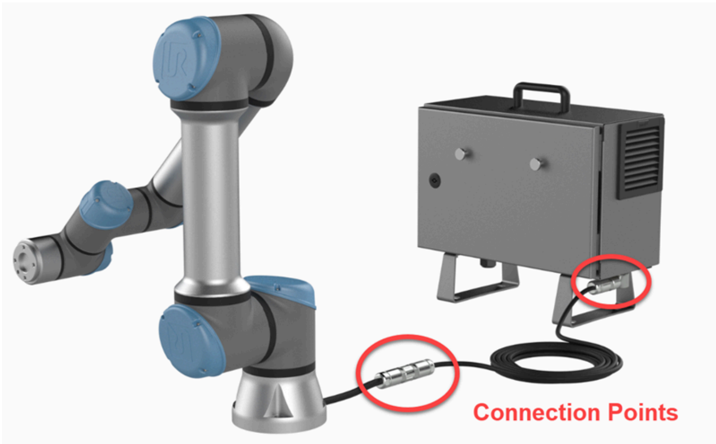
The shoulder and elbow mainly provide larger arm movements, while the wrist joints provide finer positioning and orientation.

2. Control Box
The Control Box contains the robot controller and electrical system. It supplies power to the robot and coordinates the motion of the robot joints.

3. Tool Flange
The tool flange is located at the end of the robot arm. An end-effector such as a **gripper, camera, or other tool** can be mounted here depending on the application.

4. Teach Pendant
The Teach Pendant is the main interface for operating and programming the UR5e. It runs the **PolyScope** interface and is used for robot initialization, manual movement, program configuration, and execution.





The workflow starts by setting up the environment and robot connection, followed by checking the camera. After that, collect demonstration data through teleoperation, preprocess the recorded data, train the ACT policy, and finally run the trained policy on the UR5e robot.

UR5e Operation Guide

1. Setup

Prepare the software environment and project directory required to run the UR5e experiments.

Activate Environment

```
source ~/muhammad-mujoco/bin/activate
```

Go to the project folder

```
cd /home/muhammad-ra/Desktop/AIST-ART/General\
Robot/MultimodalRobotModel/multimodal_robot_model
```

Repository
https://github.com/muhamuttaqien/MultimodalRobotModel/tree/master

2. Robot Network

Before starting, make sure the robot is properly installed and the workspace is clear.

1. **Press the Emergency Stop** button on the Teach Pendant.
2. **Press the Power button** on the Teach Pendant and wait for PolyScope to start.
3. When prompted, open the **Initialize** screen.
4. **Release the Emergency Stop** by twisting the button.
5. Make sure nobody is inside the robot workspace.
6. On the Initialize screen, press **ON** to power the robot arm.
7. Verify that the **payload and robot mounting configuration** are correct.
8. Press **START** to release the joint brakes.
9. Wait until the robot status becomes **Normal / green**.

During brake release, the robot may make clicking sounds and move slightly. This is normal. The UR5e is now powered on and ready for operation.

3. Robot Network

Configure the UR5e and control PC network settings so both devices can communicate correctly.

UR5e
IP: 192.168.11.4
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.1

PC
IP: 192.168.11.xxx
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.1

Ubuntu Network Setup
https://help.ubuntu.com/stable/ubuntu-help/net-fixed-ip-address.html.en

4. Camera Setup

Check the Intel RealSense camera connection and confirm that the camera stream is working before starting experiments. Check the Intel RealSense camera: realsense-viewer

References:
https://www.intelrealsense.com/get-started-depth-camera/
https://github.com/IntelRealSense/librealsense/blob/development/doc/distribution_linux.md
https://www.youtube.com/watch?v=zbTtNsWz6Ec

5. Gello Setup

Configure the GELLO software path so the teleoperation scripts can properly access the required GELLO modules. If needed, add the GELLO software path:

```
import os
import sys

gello_path = os.path.abspath(
'/home/muhammad-ra/Desktop/AIST-ART/General
Robot/MultimodalRobotModel/third_party/gello_software')

sys.path.append(gello_path)
```

6. Collect Demonstration Data

Use teleoperation to control the UR5e and record demonstration trajectories for training the robot policy.

Simulation

```
python teleop/bin/TeleopMujocoUR5eStore.py --world_idx_list 5
```

Reference for changing object position
https://chatgpt.com/c/67623042-cb20-8013-9552-cd300caab4f9

Real UR5e

```
python teleop/bin/TeleopRealUR5eGear.py --world_idx_list 0
```

7. Preprocess Dataset

Convert the recorded teleoperation data into the training-ready dataset required by the learning pipeline.

```
python ../utils/make_dataset.py \
--in_dir ./data/teleop_data_store_reals9_diff/ \
--out_dir ./data/learning_data_store_reals9_diff/ \
--train_ratio 0.8 \
--nproc `nproc` \
--skip 3
```

8. Train ACT

Train the ACT policy using the processed demonstration dataset and save the resulting model checkpoints.

```
python ./bin/train.py \
--dataset_dir ./data/learning_data_store_reals9_diff/ \
--ckpt_dir ./log/20250310_envreals9_diff \
--task_name sim_ur5estore \
--policy_class ACT \
--kl_weight 10 \
--chunk_size 100 \
--hidden_dim 512 \
--batch_size 2 \
--dim_feedforward 3200 \
--num_epochs 1600 \
--lr 1e-5 \
--seed 0
```

9. Run Policy

Load the trained ACT checkpoint and execute the learned policy on the real UR5e robot for evaluation.

```
python ./bin/rollout/RolloutActRealUR5eGear.py \
--ckpt_dir ./log/20250310_envreals9_diff/ \
--ckpt_name policy_last.ckpt \
--chunk_size 100 \
--seed 0 \
--skip 3 \
--world_idx 0
```

Important Scripts

/home/aist-art/Desktop/Projects/AIST-ART/General Robot/MultimodalRobotModel/multimodal_robot_model/**act/bin/train.py**

/home/aist-art/Desktop/Projects/AIST-ART/General Robot/MultimodalRobotModel/third_party/**act/policy.py**

/home/aist-art/Desktop/Projects/AIST-ART/General Robot/MultimodalRobotModel/third_party/**act/detr/main.py**

/home/aist-art/Desktop/Projects/AIST-ART/General Robot/MultimodalRobotModel/third_party/**act/detr/models/__init__.py**

/home/aist-art/Desktop/Projects/AIST-ART/General Robot/MultimodalRobotModel/third_party/**act/detr/models/detr_vae.py**

/home/aist-art/Desktop/Projects/AIST-ART/General Robot/MultimodalRobotModel/third_party/**act/detr/models/transformer.py**
https://pytorch.org/docs/stable/generated/torch.nn.MultiheadAttention.html